

23/07.

Exercise - 01

Basic Networking Commands

1. arp -a :

arp is short form of address resolution protocol, it will show the IP address of your computer along with IP address and MAC address of your computer.

Output:

Interface : 172.16.110.112 --- 0x5

Internal Address	Physical Address	Type
172.16.8.1	7c-5a-1c-cf-6e-45	dynamic
172.16.8.3	d4-3d-7e-c7-b5-3c	dynamic
172.16.8.4	d4-3d-7e-ad-3c-bf	dynamic

2. hostname :

This is the simplest of all TCP/IP commands. It simply displays the name of your computer.

Output :

DESKTOP - 20V704J

3. ipconfig /all :

This command displays detailed configuration info about your TCP/IP connection including Router, Gateway, DNS, DHCP and type of Ethernet adapter in your system.

Output :

Windows IP Configuration

Hostname : DESKTOP - 20V704J
Primary Dns Suffix :
Node Type : Hybrid
IP Routing Enabled : No
WINS Proxy Enabled : No

Ethernet adapter Ethernet 2 :

Connection - specific DNS Suffix :
Description : Intel(R) Ethernet (1)
Physical Address : D8-BB-C1-C5-CE-58
DHCP Enabled : NO

4. nbstat -a :

This command ~~displays detailed configuration information~~ ~~a~~ helps solve problems with NetBIOS name resolution.

Output :

NBSTAT [[-a RemoteName] [-A IP address] [-C] [-n]
[-r] [-R] [-RR] [-S] [-S] [interval]]

- a (adapter status)
- A (Adapter status)
- C (cache)
- n (names)
- r (resolved)
- R (Reload)
- S (Sessions)
- S (Sessions)
- RR (Release Refresh)

5. netstat :

displays a variety of statistics about a computer's active TCP / IP connections.

Output :

Active connections

Proto	Local Address	Foreign Address	State
TCP	172.16.10.112:7680	172.16.10.102:53514	Established
TCP	172.16.10.112:51297	4.213.25.240:https	Established
TCP	172.16.10.112:51908	172.16.8.127:ssh	Time wait
TCP	172.16.10.112:51946	10bomp-in-f188:5228	Established

6. nslookup :

It is a tool used to perform DNS lookup in Linux. It is used to display DNS details, such as IP address of a particular computer, the MX records for a domain or the NS server of a domain.

Output :

Default Server : Unknown

Address : 172.16.8.1

7. pathping :

pathping is unique to Windows, and is basically a combination of the ping and Tracert commands. Pathping traces the route to the destination address then launches a 25 second test of each router along the way.

Output :

usage : pathping [-q ~~host~~ list] [-n maximum_hops] [-i address]
[-p period] [-q num-queries] [-w timeout]
[-4] [-6] target-name.

8. Ping hostname :

this command is the best way to test connectivity between two nodes.

Output :

PING localhost.localdomain (localhost (:::1)) 56 data bytes

64 bytes from localhost (:::1) : icmp_seq=1 ttl=64

64 bytes from localhost (:::1) : icmp_seq=2 ttl=64

64 bytes from localhost (:::1) : icmp_seq=3 ttl=64

64 bytes from localhost (:::1) : icmp_seq=4 ttl=64

9. route :

route command is used to show / manipulate the IP routing table. It is primarily used to setup static route to specific host or networks via an interface.

Output:

Route [-f] [-P] [-A] [-G] command [destination]
[MASK netmask] [gateway] [METRIC metric]

Command One of these:

PRINT Prints a route

ADD Adds a route

DELETE Deletes a route

CHANGE Modifies existing route.

Linux Networking Commands :

1. ifconfig

Output:

1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc
nonqueue state UNKNOWN group default qlen 1000

```
link / loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
    valid-lft forever preferred-lft forever
inet ::1/128 scope host
```

2. ifconfig

Output:

```
enp3s0: flags=4163<UP, BROADCAST, RUNNING, MULTICAST>
    inet 172.16.8.127 netmask 255.255.252.0
    broadcast 172.16.11.255
    inet6 fe80::a6fb:cd5f:131c:6e50
    ether 00:27:0e:13:f6:d9
```

3. mtr

Output:

My traceroute [V0.87]

localhost.localdomain (0.0.0.0) Thu Jul 23 09:36:58 2026

Keys:	Help	Display mode	Restart	Statistics	order of fields
		loss %	int	last	avg best worst dev
localhost		0.0%	128	0.1	0.2 0.1 1.9 0.2

4. tcpdump

Output:

tcpdump: you don't have permission to capture on that device.

Result:

Thus the basic networking and linux commands are studied and learned successfully.